

3 Ans: C

Nature of collision is inelastic, hence total KE is not conserved.

By C.O.M.

$$mu_x = 2mv_{xy} \rightarrow v_{xy} = mu_x/2m = \frac{1}{2} u_x$$

$$\text{initial KE} = \frac{1}{2} mu_x^2 = E_k =$$

$$\text{loss KE} = \frac{1}{2} mu_x^2 - \frac{1}{2} mv_{xy}^2 = E_k - \frac{1}{2} (2m)(\frac{1}{2} u_x)^2 = E_k - \frac{1}{2} E_k = \frac{1}{2} E_k$$